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Assignment-5

1) $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x|x| \rightarrow$ inflection pt at $x=0$.

Proof: Let $\delta=1$. $\forall x \in (0, \delta)$ ~~or~~ $(0, 1)$, we have,

$f(x) = x^2$, which is differentiable on $(0, 1)$.

$f'(x) = 2x$ which is increasing on $(0, 1)$. Hence f is convex in $(0, 1)$.

$\forall x \in (-\delta, 0)$ or $(-1, 0)$ we have $f(x) = -x^2$.

f is differentiable on $(-1, 0)$ and $f'(x) = -2x$ is decreasing on $(-1, 0)$.

$\therefore f$ is concave on $(-1, 0)$.

Hence 0 is an inflection point of f as f changes concavity at $x=0$.

2) (a) $f: (a, b) \rightarrow \mathbb{R}$, diff. Then (i) f convex $\Leftrightarrow f'$ increasing.

Solⁿ: Suppose f is convex. Consider $x_1, x_2 \in (a, b)$ such that $x_1 < x_2$. Let $x_1 < u < v < x_2$.

Then

$$\frac{f(u) - f(x_1)}{u - x_1} \leq \frac{f(v) - f(x_2)}{v - x_2}$$

$$\Rightarrow \lim_{u \rightarrow x_1^+} \frac{f(u) - f(x_1)}{u - x_1} \leq \lim_{v \rightarrow x_2^-} \frac{f(v) - f(x_2)}{v - x_2}$$

$$\Rightarrow f'(x_1) \leq f'(x_2) \text{ for } x_1 < x_2.$$

Hence f' is increasing on (a, b) .

Now if f' is increasing on (a, b) we must prove convexity.

Let $x_1, x_2 \in (a, b)$, $x_1 < x_2$.

Let $x_t := tx_1 + (1-t)x_2$ for some $t \in (0, 1)$.

Then $x_1 < x_t < x_2$. continuous and $f: (x_1, x_2) \rightarrow \mathbb{R}$ is diff.
 $f: [x_1, x_2] \rightarrow \mathbb{R}$ is differentiable so by LMVT,
 $\exists c_1 \in (x_1, x_t)$ such that

$$f'(c_1) = \frac{f(x_t) - f(x_1)}{x_t - x_1}$$

$$x_t - x_1 = (1-t)(x_2 - x_1)$$

$$\therefore f'(c_1) = \frac{f(x_t) - f(x_1)}{(1-t)(x_2 - x_1)}$$

Similarly $\exists c_2 \in (x_t, x_2)$ such that

$$f'(c_2) = \frac{f(x_2) - f(x_t)}{x_2 - x_t}$$

$$\begin{aligned} x_2 - x_t &= x_2 - tx_1 - (1-t)x_2 \\ &= t(x_2 - x_1) \end{aligned}$$

$$\therefore f'(c_2) = \frac{f(x_2) - f(x_t)}{t(x_2 - x_1)}$$

Since f' is increasing we have,

$$\begin{aligned} f'(c_1) &\leq f'(c_2) \\ \frac{f(x_t) - f(x_1)}{(1-t)(x_2 - x_1)} &\leq \frac{f(x_2) - f(x_t)}{t(x_2 - x_1)} \end{aligned}$$

$$tf(x_t) - tf(x_1) \leq (1-t)f(x_2) - f(x_t) + tf(x_2)$$

$$f(x_t) \leq (1-t)f(x_2) + tf(x_1), \quad \forall t \in (0,1).$$

Hence by definition f is convex.

(b) f is twice differentiable. Then f is convex iff $f''(x) \geq 0$.

Solⁿ: Suppose f is convex. Then we know that f' is increasing on (a, b) .
Since f' is increasing, $f''(x) \geq 0$ for $x \in (a, b)$.

($\Leftarrow$)
Now suppose $f''(x) \geq 0$ for $x \in (a, b)$.

$\Rightarrow f'$ is increasing on (a, b) .

$\Rightarrow f$ is convex on (a, b) .

3. Show that a deg 4 poly. has either 0 or 2 inflection pts.

Solⁿ: ~~Since~~ Let $P(x)$ be a 4th degree polynomial. Since P is twice differentiable on $\mathbb{R}$, $f''(c) = 0$ is necessary for c to be a point of inflection.

P must have less than 3 inflection points since $P''(x)$ is of degree 2 so it has at most 2 roots.

Suppose P'' has two distinct roots α, β . Then $P''(x) = k(x - \alpha)(x - \beta)$ for some $k \in \mathbb{R}$.

$$P'''(x) = k(2x - \alpha - \beta)$$

$$P'''(\alpha) = k(\alpha - \beta) \neq 0$$

$$P'''(\beta) = k(\beta - \alpha) \neq 0$$

Hence α, β are inflection points of P .

Suppose P'' has 1 root with multiplicity 2.

Then $P''(x) = k(x - \alpha)^2$ for some $k, \alpha \in \mathbb{R}$.

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4) Darboux theorem.

$f: [a, b] \rightarrow \mathbb{R}$ cts & diff. $f'(a) \neq f'(b)$. $\Rightarrow$ b/w $f'(a)$ & $f'(b)$. Find p.t. $\exists c \in (a, b)$ s.t. $f'(c) = \lambda$.

Let $g(t) := f(t) - \lambda t$; $g: [a, b] \rightarrow \mathbb{R}$.

Then $g'(t) = f'(t) - \lambda$

Since g is cts on $[a, b]$, by min-max theorem it attains a min & max on $[a, b]$.

Suppose wlog that $f'(a) < \lambda < f'(b)$.

Let c be a point of minimum of g .

We know $g'(a) = f'(a) - \lambda < 0$. So $\exists \delta_1 > 0$ st $\forall x \in (a, a + \delta_1)$ we have $g(x) < g(a)$.

Similarly $\exists \delta_2 > 0$ such that $\forall x \in (b - \delta_2, b)$ we have $g(x) < g(b)$.

$\therefore$ Neither a nor b can be points of minimum. So $c \in (a, b)$.

Hence by Local-Minimum theorem $g'(c) = 0$.

$$f'(c) - \lambda = 0$$

$$\boxed{f'(c) = \lambda}$$

Hence proved

5) Compute Taylor Poly $P_n(x)$ for $f: (-1, \infty) \rightarrow \mathbb{R}$ given by $f(x) = \ln(1+x)$ at $a=0$. Use Taylor's theorem to conclude that for each $n \geq 1$ we have $P_{2n}(x) < f(x) < P_{2n+1}(x)$ for $x > 0$. Is f analytic at $x=0$?

Solⁿ: $f(x) = \ln(1+x)$.

Claim: $f^{(n+1)}(x) = (-1)^n n! (1+x)^{-(n+1)} \quad \forall n \in \mathbb{N} \cup \{0\}$

Proof: We shall prove it by induction.

Base case: $n=0$

$$f'(x) = \frac{1}{1+x}$$

$$f(x) = \ln(1+x) \Rightarrow f'(x) = \frac{1}{1+x}$$

$$f'(x) = (1+x)^{-1} \\ = (-1)^0 (0!) (1+x)^{-(0+1)}$$

Hence the base case is true.

Suppose the claim is true for $n=k$.

Let $g(x) = f^{(k+1)}(x)$.

Then $f^{(k+2)}(x) = g'(x)$

$$g'(x) = \frac{(-1)^k k! (-k-1) \cdot (1+x)^{-k-2}}{(1+x)^{-k-2}} = \frac{(-1)^{k+1} (k+1)! (1+x)^{-(k+2)}}{(1+x)^{-(k+2)}} \\ = (-1)^{k+1} (k+1)! (1+x)^{-(k+2)}$$

Hence by induction the claim is true for all $n \in \mathbb{N} \cup \{0\}$.

$$P_n = f(0) + \sum_{k=1}^n \frac{f^{(k)}(0)}{k!} \cdot (x^k)$$

$$= \sum_{k=1}^n \frac{(-1)^{k-1} (k-1)! \cdot x^k}{k!} = \sum_{k=1}^n \frac{(-1)^{k-1} \cdot x^k}{k}$$

$$= x - \frac{x^2}{2} + \dots + \frac{x^n (-1)^{n+1}}{n}$$

Now we shall prove $P_n(x) < f(x) < P_{n+1}(x)$ for $x > 0$.

We know $f(x) = P_n(x) + R_{n+1}(x)$.

$$R_{n+1}(x) = \frac{f^{(n+1)}(c)}{(n+1)!} x^{n+1} \quad \text{where } c \in (0, x).$$

Show that $R_{2n}(x) > 0$ and $R_{2n+1}(x) < 0$ ✓.

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5. cont... $P_{2n}(x) = \frac{f^{(2n+1)}(c) x^{2n+1}}{(2n+1)!}$ where $c \in (0, x)$.

$$P_{2n}(x) = \frac{(-1)^{2n} (2n)! (1+c)^{-(2n+2)} \cdot x^{2n+1}}{(2n+1)!}$$

$$= \frac{x^{2n+1}}{(2n+1)(1+c)^{2n+1}} > 0 \quad (\because x > 0, 1+c > 0)$$

$$\therefore f(x) - P_{2n}(x) > 0 \Rightarrow P_{2n}(x) < f(x).$$

Similarly

$$P_{2n+1}(x) = \frac{-x^{2n+2}}{(2n+2)(1+c)^{2n+2}} < 0$$

$$\therefore f(x) - P_{2n+1}(x) < 0 \Rightarrow f(x) < P_{2n+1}(x).$$

$$\therefore \underline{P_{2n}(x) < f(x) < P_{2n+1}(x)} \text{ for } x > 0.$$

Now, to prove f is analytic at $x=0$.

$$|R_n(x)| = \frac{|x|^{n+1}}{(n+1)(1+c)^{n+1}}$$

To test the convergence of this, apply Ratio test. ($\because R_n(x) \neq 0$ for $x \neq 0$)

$$\left| \frac{R_{n+1}(x)}{R_n(x)} \right| = \frac{|x| \cdot n}{n+1} \rightarrow |x| \text{ as } n \rightarrow \infty$$

$\therefore R_n(x)$ converges for $|x| < 1$.

Clearly for ~~all~~, $|R_n(x)| \rightarrow 0$ as $n \rightarrow \infty$ since it is a geometric sequence with common ratio $\left| \frac{x}{1+c} \right|$.

$$\therefore \text{for } -\frac{1}{2} < x < 0 < c \Rightarrow x < c+1 \text{ so } \left| \frac{x}{c+1} \right| < 1$$

For $-\frac{1}{2} < x < 0$:

$$-1 < 2x$$

$$\Rightarrow -1 < +x + x$$

$$-n-1 < n$$

$$n+1 > -x$$

$$n+1 > |x|$$

$$\frac{|x|}{n+1} < 1$$

$$\Rightarrow \left(\frac{|x|}{n+1} \right)^{n+1} < 1$$

$$\text{And } |R_n(x)| = \frac{|x|^{n+1}}{(n+1)(1+x)^{n+1}} < \frac{|x|^{n+1}}{(n+1)(1+x)^{n+1}} < \frac{1}{n+1} \rightarrow 0 \text{ as } n \rightarrow \infty$$

$\therefore R_n(x) \rightarrow 0$ as $n \rightarrow \infty$ for $-\frac{1}{2} < x < 0$.

The Taylor series $T(x)$ is convergent for $|x| < 1$ ~~for~~ by computing radius of convergence using ratio test.

$$T(x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$$

Hence f is analytic at $x=0$.

6. a) $\lim_{n \rightarrow 0^+} x^n$

$$= \lim_{n \rightarrow 0^+} e^{n \ln x}$$

Let $g(n) = n \ln x$

Now, $n \ln x = \lim$

Let $f(n) = \ln x$, $\forall x$

$$g(n) = \frac{1}{n}$$

$$g'(n) = -\frac{1}{n^2} \neq 0 \quad \forall x > 0$$

$$f'(n) = \frac{1}{n}$$

$$\frac{f'(n)}{g'(n)} = \frac{1/n}{-1/n^2} = -n$$

$$\lim_{n \rightarrow 0^+} \frac{f'(n)}{g'(n)} = 0$$

$$\lim_{n \rightarrow 0^+} f(n) = -\infty, \quad \lim_{n \rightarrow 0^+} g(n) = \infty$$

$$\therefore \text{By L'Hôpital's rule, } \lim_{n \rightarrow 0^+} \frac{f(n)}{g(n)} = \lim_{n \rightarrow 0^+} \frac{f'(n)}{g'(n)} = 0$$

$$\therefore \lim_{n \rightarrow 0^+} n \ln n = 0.$$

Hence $\forall (n_n)$ such that $n_n \rightarrow 0$ as $n \rightarrow \infty$ and we have
 ~~$y_n = n_n \ln(n_n) \rightarrow 0$~~ as $n \rightarrow \infty$.

The function e^x is ~~con~~ $g(n) = g: [0, \infty) \rightarrow \mathbb{R}$ is ~~conti~~,
 $g(n) = e^n$ is continuous.

$$\text{Hence } g(y_n) = e^{y_n} = e^{n_n \ln n_n} \rightarrow e^0 = 1 \text{ as } n \rightarrow \infty.$$

Let $h: (0, \infty) \rightarrow \mathbb{R}$, $h(n) = e^{n \ln n}$.

Then $\lim_{n \rightarrow \infty} h(n_n) = 1 \quad \forall (n_n) \text{ s.t. } n_n \rightarrow 0 \text{ as } n \rightarrow \infty.$

$$\text{Hence } \lim_{n \rightarrow 0^+} n^n = 1.$$

$$(b) \lim_{n \rightarrow 0} \frac{\sin n}{n}$$

solⁿ: Let $f: (-1, 1) \rightarrow \mathbb{R}$, $f(n) = \sin n$
 $g: (-1, 1) \rightarrow \mathbb{R}$, $g(n) = n$.

$$\lim_{n \rightarrow 0} f(n) = 0, \quad \lim_{n \rightarrow 0} g(n) = 0.$$

$$g'(n) = 1 \neq 0 \quad \forall n \in (-1, 1).$$

Hence by L'Hôpital's rule,

$$\lim_{n \rightarrow 0} \frac{f'(n)}{g'(n)} = \lim_{n \rightarrow 0} \frac{\cos n}{1} = 1$$

Hence by L'Hôpital's rule,

$$\lim_{n \rightarrow 0} \frac{f(n)}{g(n)} = \lim_{n \rightarrow 0} \frac{f'(n)}{g'(n)} = 1.$$

$$\therefore \lim_{n \rightarrow 0} \frac{\sin n}{n} = 1.$$

c) ✓ | d) $\lim_{n \rightarrow 0} (1-n)^{1/n}$

$$= \lim_{n \rightarrow 0} e^{\frac{1}{n} \ln(1-n)}.$$

$$\lim_{n \rightarrow 0} \frac{\ln(1-n)}{n} = ?$$

Let $f: (-1, 1) \rightarrow \mathbb{R}, f(n) = \ln(1-n)$
 $g: (-1, 1) \rightarrow \mathbb{R}, g(n) = n.$

$$\lim_{n \rightarrow 0} \frac{f'(n)}{g'(n)} = \lim_{n \rightarrow 0} \frac{-\frac{1}{1-n}}{1} = -1$$

$$g'(n) = 1 \neq 0 \forall n \in (-1, 1).$$

$$\lim_{n \rightarrow 0} f(n) = \ln(1-n) = 0, \quad \lim_{n \rightarrow 0} n = 0.$$

Hence by L'Hôpital's rule,

$$\lim_{n \rightarrow 0} \frac{f(n)}{g(n)} = \lim_{n \rightarrow 0} \frac{f'(n)}{g'(n)} = -1.$$

$$\therefore \lim_{n \rightarrow 0} e^{\frac{1}{n} \ln(1-n)} = e^{\lim_{n \rightarrow 0} \frac{1}{n} \ln(1-n)} \quad (\because e^n \text{ is continuous } \forall n \in \mathbb{R})$$

$$= e^{-1} = \left(\frac{1}{e}\right)$$

7. P.T $E_n(n) \rightarrow 0$ as $n \rightarrow \infty$ for $n \geq 0$ for e^n at $n=0$.
 $[0, \infty)$

Solⁿ: Let $f: \mathbb{R} \rightarrow \mathbb{R}, f(n) = e^n$.

Let $T_n(n)$ be the Taylor polynomial of f of n^{th} degree.

By Taylor's theorem,

$$f(x) = T_n(x) + E_n(x) \quad \forall x \in \mathbb{R} \quad \forall n > 0.$$

$$E_n(x) = \frac{f^{(n+1)}(c) \cdot x^{n+1}}{(n+1)!} \quad \text{for some } c \text{ in } (0, x).$$

$$0 < c < x$$

$$f^{(n+1)}(x) = e^x$$

$$\therefore E_n(x) = \frac{e^c \cdot x^{n+1}}{(n+1)!} < \frac{e^x \cdot x^{n+1}}{(n+1)!} \quad (\because 0 < c < x)$$

$$E_n(x) \neq 0 \quad \forall n \in \mathbb{N} \text{ since } x > 0.$$

\(\therefore\) Applying ratio test,

$$\left| \frac{E_{n+1}(x)}{E_n(x)} \right| = \left| \frac{e^c \cdot x^{n+2} \cdot (n+1)!}{(n+2)! \cdot e^c \cdot x^{n+1}} \right|$$

$$= \frac{|x|}{n+2} \rightarrow 0 < 1 \text{ as } n \rightarrow \infty.$$

Hence $E_n(x)$ converges to 0 as $n \rightarrow \infty$.

8. Compute Taylor series of $\sin$ and $\cos$ at $x = 0, \frac{\pi}{2}, -\frac{\pi}{2}$.

Find their radius of convergence.

For $\delta = 0.1$ and tolerance $\epsilon = 10^{-10}$, compute the smallest degree n of the Taylor poly. s.t. $|\sin x - p_n(x)| < \epsilon$ $\forall x \in (-\delta, \delta)$.

Solⁿ: $\sin$ at $x=0$

$$f: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}, f(x) = \sin x.$$

$$f^{(4k)}(0) = 0 \quad f^{(4k)}(x) = \sin x \quad \forall k \in \mathbb{N} \cup \{0\}.$$

$$f^{(4k+1)}(x) = \cos x$$

$$f^{(4k+2)}(x) = -\sin x, \quad f^{(4k+3)}(x) = -\cos x$$

$$\begin{aligned} \therefore f^{(4k)}(0) &= 0 \\ f^{(4k+1)}(0) &= 1 \\ f^{(4k+2)}(0) &= 0 \\ f^{(4k+3)}(0) &= -1 \quad \forall k \in \mathbb{N} \cup \{0\} \end{aligned}$$

$$\begin{aligned} \therefore T_{\bullet}(x) &= \sum_{k=1}^{\infty} \frac{f^{(k)}(0) x^k}{k!} \\ &= x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \\ &= \sum_{k=1}^{\infty} \frac{(-1)^{k+1} \cdot x^{2k-1}}{(2k-1)!} \end{aligned}$$

$$\text{Let } a_k := \frac{(-1)^{k+1} \cdot x^{2k-1}}{(2k-1)!}$$

For $x \neq 0$, $a_k \neq 0 \quad \forall k \in \mathbb{N}$

$\therefore$ Applying ratio test,

$$\left| \frac{a_{k+1}}{a_k} \right| = \frac{|x^2|}{(2k)(2k+1)} \rightarrow 0 \text{ as } k \rightarrow \infty \quad \forall x \in \mathbb{R} \setminus \{0\}.$$

Hence the radius of convergence

Hence the Taylor series converges $\forall x \in \mathbb{R}$

The error term is

$$R_n(x) = \frac{f^{(n+1)}(c) x^{n+1}}{(n+1)!} \quad \text{for some } c \text{ between } 0 \text{ and } x$$

~~Let~~

$$f^{(n)}(x) \in [-1, 1] \quad \forall n \in \mathbb{N}, x \in \mathbb{R}.$$

$$\therefore |f^{(n+1)}(c)| \leq 1$$

$$|R_n(x)| \leq \frac{|x|^{n+1}}{(n+1)!} \rightarrow 0 \text{ as } n \rightarrow \infty \quad \forall x \in \mathbb{R}$$

Hence the radius of convergence of the Taylor series is ∞ .

Now, $\delta = 0.1$, $\epsilon = 10^{-10}$

$$|S_{n+1} - P_n(x)| = |R_n(x)| < \epsilon \quad \forall x \in (-\delta, \delta).$$

$$\Rightarrow \frac{|f^{(n+1)}(c) x^{n+1}|}{(n+1)!} < \epsilon \quad \text{for where } c \text{ is between } 0 \text{ and } x.$$

$$|x| < \delta = 0.1, \quad |f^{(n+1)}(c)| \leq 1$$

$$\therefore \frac{(0.1)^{n+1}}{(n+1)!} < 10^{-10}$$

$$\frac{10^{-(n+1)}}{(n+1)!} < 10^{-10}$$

$$n=6$$

9. $f(x) = x|x|$ is differentiable once but not twice.

$$f(x) = \begin{cases} x^2, & x \geq 0 \\ -x^2, & x < 0 \end{cases}$$

$$f'(x) = \begin{cases} 2x & \text{for } x \geq 0 \\ -2x & \text{for } x < 0 \end{cases}$$

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \\ = \lim_{h \rightarrow 0} \frac{f(h)}{h}$$

But $f''(x)$ is not defined at $x=0$.

$$\text{for } h \geq 0: \quad \lim_{h \rightarrow 0^+} \frac{f(h)}{h} = \lim_{h \rightarrow 0^+} \frac{h^2}{h} = 0$$

$$f''(0) = \lim_{h \rightarrow 0} \frac{f'(h) - f'(0)}{h}$$

$$\lim_{h \rightarrow 0^-} \frac{f(h)}{h} = \lim_{h \rightarrow 0^-} \frac{-h^2}{h} = 0$$

$$= \lim_{h \rightarrow 0} \frac{f'(h)}{h} \text{ if exists}$$

For $h \geq 0$:

$$\lim_{h \rightarrow 0^+} \frac{f'(h)}{h} = \lim_{h \rightarrow 0^+} \frac{2h}{h} = 2$$

$\therefore \lim_{h \rightarrow 0} \frac{f'(h)}{h}$ doesn't exist.

for $h < 0$:

$$\lim_{h \rightarrow 0^-} \frac{f'(h)}{h} = \lim_{h \rightarrow 0^-} \frac{-2h}{h} = -2$$

f is not twice differentiable.

In general: $f_n(x) = x^n |x|$ is n times differentiable but not $n+1$ times.

10. $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) := \begin{cases} e^{-\frac{1}{x^2}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ smooth.

① $\lim_{n \rightarrow 0} \frac{f(n)}{n} = L$ iff $\lim_{n \rightarrow \infty} n f\left(\frac{1}{n}\right) = \lim_{n \rightarrow -\infty} n f\left(\frac{1}{n}\right) = L$.

Solⁿ: Given, $\lim_{n \rightarrow 0} \frac{f(n)}{n} = L$.

Let (x_n) be a sequence such that $x_n \rightarrow \infty$. Then $\frac{1}{x_n} \rightarrow 0$. Hence

$$\lim_{n \rightarrow \infty} \frac{f\left(\frac{1}{x_n}\right)}{\frac{1}{x_n}} = \lim_{n \rightarrow \infty} x_n f\left(\frac{1}{x_n}\right) = L.$$

$\therefore \lim_{n \rightarrow \infty} n f\left(\frac{1}{n}\right) = L$.

Let (y_n) be a sequence such that $y_n \rightarrow -\infty$ as $n \rightarrow \infty$. Then $\frac{1}{y_n} \rightarrow 0$ so $\lim_{n \rightarrow \infty} y_n f\left(\frac{1}{y_n}\right) = L$.

Hence $\lim_{n \rightarrow -\infty} n f\left(\frac{1}{n}\right) = L$.

($\Leftarrow$)

Now suppose we know $\lim_{n \rightarrow \infty} n f\left(\frac{1}{n}\right) = \lim_{n \rightarrow -\infty} n f\left(\frac{1}{n}\right) = L$.

$\therefore \forall \epsilon > 0, \exists \delta_1 > 0$ such that $\forall x > \delta_1, 0 < x f\left(\frac{1}{x}\right) - L < \epsilon$ and

$\exists \delta_2 > 0$ s.t. $\forall x < -\delta_2, x f\left(\frac{1}{x}\right) - L < \epsilon$.

$x > \delta_1 \Leftrightarrow \frac{1}{x} < \frac{1}{\delta_1}$ and $x < -\delta_2 \Leftrightarrow \frac{1}{x} > -\frac{1}{\delta_2}$

$\therefore \frac{1}{\delta_2} < \frac{1}{x} < \frac{1}{\delta_1}$

Let $y := \frac{1}{n}$ and $\delta := \min \left\{ \left| \frac{1}{\delta_1} \right|, \left| \frac{1}{\delta_2} \right| \right\}$.

Then $\forall |y| < \delta$ we have $\left| x f\left(\frac{1}{n}\right) - L \right| = \left| \frac{f(y)}{y} - L \right| < \epsilon$.

Hence $\lim_{y \rightarrow 0} \frac{f(y)}{y} = L$.

⑦ Compute $\lim_{n \rightarrow \infty} x f\left(\frac{1}{n}\right)$.

Solⁿ: $\lim_{n \rightarrow \infty} x f\left(\frac{1}{n}\right) = x \cdot e^{-x^2} \lim_{n \rightarrow \infty} x e^{-x^2}$

$$= \lim_{n \rightarrow \infty} \frac{x}{e^{x^2}}$$

Let $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x$ and $g: \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = e^{x^2}$.
 $g'(x) = 2xe^{x^2} \neq 0 \quad \forall x > 0$.

$$\frac{f'(x)}{g'(x)} = \frac{1}{2xe^{x^2}}$$

$$\lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = 0$$

$$\therefore \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} x = \infty$$

$$\lim_{x \rightarrow \infty} g(x) = \lim_{x \rightarrow \infty} e^{x^2} = \infty$$

$\therefore$ By L'Hôpital's rule, $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = 0$.

⑧ P.T. $e^{-1/x^2} = o(x^m) \quad \forall m > 0$.

Proof: First we shall prove it for $m \in \mathbb{N}$, by induction.

Base case: $m=1$, $m=2$.

Let $f(x) = e^{-1/x^2}$, $g(x) = x^{m-1}$, $g: \mathbb{R} \rightarrow \mathbb{R}$.

$$\Rightarrow g\left(\frac{1}{n}\right) = \frac{e^{-n^2}}{\left(\frac{1}{n}\right)^{m-1}} = \frac{n^{m-1}}{e^{n^2}}$$

We need to compute $\lim_{n \rightarrow \infty} \frac{e^{-1/n^2}}{n^m} = \lim_{n \rightarrow \infty} \frac{g(n)}{n}$.

By (I), this is equal to

$$\lim_{x \rightarrow 0} x g\left(\frac{1}{x}\right) = \lim_{n \rightarrow 0} \frac{x^n}{e^{x^2}}$$

For $m=1$:

$$\lim_{x \rightarrow 0} \frac{x}{e^{x^2}} = \lim_{x \rightarrow 0} \frac{e^{-x^2}}{x} = 0 \quad (\text{By (I)})$$

For $m=2$:

$$\lim_{x \rightarrow 0} \frac{x^2}{e^{x^2}} = 0$$

~~We know Let $P(x) = x^2$, $Q(x) = e^{x^2}$,
 Let $P: \mathbb{R} \rightarrow \mathbb{R}$, $Q: \mathbb{R} \rightarrow \mathbb{R}$ be $P(x) := x^2$, $Q(x) := e^{x^2}$.
 $P'(x) = 2x$
 $Q'(x) = 2xe^{x^2} \neq 0 \quad \forall x \neq 0$.
 $\lim_{x \rightarrow 0} \frac{P'(x)}{Q'(x)} = \lim_{x \rightarrow 0} \frac{2x}{2xe^{x^2}} = \lim_{x \rightarrow 0} \frac{1}{e^{x^2}} =$~~

Induction hypothesis: Suppose $\lim_{n \rightarrow 0} \frac{x^m}{e^{x^2}} = 0$

$\forall m \leq k$, where $k \in \mathbb{N}$.

we need to find

Then $\lim_{x \rightarrow 0} \frac{x^{m+1}}{e^{x^2}}$

Let $P, Q: \mathbb{R} \rightarrow \mathbb{R}$ be $P(x) := x^{m+1}$, $Q(x) := e^{x^2}$

$$P'(x) = (m+1)x^m, \quad Q'(x) = 2xe^{x^2}$$

$$Q'(x) \neq 0 \quad \forall x \neq 0$$

$$\lim_{x \rightarrow 0} \frac{P'(x)}{Q'(x)} = \lim_{x \rightarrow 0} \frac{(m+1)x^m}{2xe^{x^2}} = \lim_{x \rightarrow 0} \frac{(m+1)}{2} \frac{x^{m-1}}{e^{x^2}}$$

By the induc

$$= 0 \quad (\text{By induction hypothesis})$$

Hence $\lim_{x \rightarrow 0} \frac{e^{-x^2}}{x^m} = 0 \quad \forall m \in \mathbb{N}$

Now suppose $m \in \mathbb{R}, m > 1$.

Then by A.P $\exists n \in \mathbb{N}$ such that $n \leq m < n+1$.

$$\frac{e^{-1/x^2}}{|x|^n} > \frac{e^{-1/x^2}}{|n|^m} > \frac{e^{-1/n^2}}{|n|^{n+1}}$$

For $0 < |x| < 1$,

$$\frac{e^{-1/x^2}}{|x|^n} > \frac{e^{-1/x^2}}{|n|^m} > \frac{e^{-1/n^2}}{|n|^{n+1}}$$

By S.A.T we know $\lim_{x \rightarrow 0} \frac{e^{-1/x^2}}{|x|^n} = \lim_{n \rightarrow \infty} \frac{e^{-1/n^2}}{|n|^{n+1}} = 0$.

By Sandwich theorem,

$$\lim_{x \rightarrow 0} \left| \frac{e^{-1/x^2}}{x^m} \right| = 0 \Rightarrow \lim_{x \rightarrow 0} \frac{e^{-1/x^2}}{x^m} = 0.$$

Hence we proved the claim $\forall m > 1$.

If $m \in (0, 1)$ then for $|x| < 1$ we have

$$|x|^m > |x|.$$

$$\therefore 0 < \frac{e^{-1/x^2}}{|x|^m} < \frac{e^{-1/x^2}}{|x|}.$$

By Sandwich theorem, $\lim_{x \rightarrow 0} \frac{e^{-1/x^2}}{x^m} = 0$.

Hence for all $m > 1$, $e^{-1/x^2} = o(x^m)$.

IV) P.T there exist polynomials $P_n(x), Q_n(x)$ s.t. $f^{(n)}(x) = \frac{P_n(x)}{Q_n(x)} e^{-1/x^2}$.

V) Finish.

$$\lim_{x \rightarrow 0} f^{(n)}(x) = \lim_{x \rightarrow 0} \frac{P_n(x)}{Q_n(x)} \cdot e^{-1/x^2}$$

$$\text{Let } Q_n(x) = a_k x^k + a_{k-1} x^{k-1} + \dots + a_1 x + a_0.$$

Let m be the smallest integer such that $a_m \neq 0$.

$$\text{Then } Q_n(x) = x^m (a_k x^{k-m} + a_{k-1} x^{k-m-1} + \dots + a_m).$$

$$\lim_{x \rightarrow 0} f^{(n)}(x) = \lim_{x \rightarrow 0} \frac{P_n(x)}{x^m} \cdot \frac{e^{-1/x^2}}{a_k x^{k-m} + \dots + a_m} = 0 \cdot \text{const} = 0$$